

## KAMALASAI, Thailand

WMO: 483900

Lat: 16.3319N

Lon: 103.5875E

Elev: 141

StdP: 99.64

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 13.6       | 15.3 | 8.8                           | 7.1 | 17.3 | 10.3 | 7.9 | 18.2 | 8.0                   | 22.8 | 6.4 | 22.8 | 0.9                   | N/A  | 0.342 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 10.6                   | 37.9            | 26.4 | 36.7 | 26.3 | 35.6 | 26.2 | 28.2                | 32.9 | 27.9 | 32.6 | 27.6 | 32.2 | 2.8                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.2                            | 23.4 | 30.2 | 26.9 | 22.9 | 29.9 | 26.6 | 22.5 | 29.6 | 92.2          | 33.2 | 90.5 | 32.6 | 89.1 | 32.3 | 31.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 6.3               | 5.5  | 5.0 | 10.7                       | 40.0 | 1.6                | 1.1 | 9.5                                                | 40.8 | 8.6        | 41.4 | 7.7        | 42.0 | 6.5        | 42.8 |    |    |
|                   |      |     | 9.1                        | 29.4 | 1.5                | 0.9 | 8.0                                                | 30.1 | 7.2        | 30.6 | 6.4        | 31.2 | 5.3        | 31.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.5   | 23.8  | 25.5  | 28.6  | 30.4  | 30.1  | 29.5  | 28.6  | 28.3  | 28.0  | 27.5  | 26.0  | 23.8 |
|                                                                                  | DBStd       | 2.98   | 2.61  | 3.06  | 2.35  | 2.18  | 1.58  | 1.45  | 1.79  | 1.20  | 1.14  | 1.31  | 1.87  | 2.81 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 4      | 2     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                                  | CDD10.0     | 6394   | 428   | 434   | 576   | 611   | 623   | 586   | 578   | 567   | 541   | 543   | 479   | 427  |
|                                                                                  | CDD18.3     | 3356   | 171   | 202   | 318   | 361   | 365   | 336   | 319   | 309   | 291   | 285   | 229   | 169  |
|                                                                                  | CDH23.3     | 36220  | 1461  | 1915  | 3579  | 4561  | 4484  | 4053  | 3562  | 3322  | 2999  | 2833  | 2036  | 1414 |
|                                                                                  | CDH26.7     | 15561  | 537   | 843   | 1803  | 2473  | 2217  | 1822  | 1403  | 1158  | 981   | 1049  | 771   | 506  |
| Wind                                                                             | WSAvg       | 1.8    | 2.1   | 2.1   | 1.8   | 1.5   | 1.7   | 1.8   | 1.9   | 1.8   | 1.5   | 1.6   | 1.9   | 2.2  |
| Precipitation                                                                    | PrecAvg     | 1403   | 9     | 18    | 43    | 77    | 177   | 201   | 228   | 269   | 250   | 107   | 21    | 2    |
|                                                                                  | PrecMax     | 1712   | 81    | 87    | 113   | 178   | 298   | 337   | 518   | 480   | 419   | 298   | 76    | 11   |
|                                                                                  | PrecMin     | 1135   | 0     | 0     | 2     | 11    | 46    | 69    | 80    | 76    | 79    | 10    | 0     | 0    |
|                                                                                  | PrecStd     | 146    | 18    | 27    | 34    | 46    | 66    | 75    | 95    | 96    | 89    | 78    | 23    | 4    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 33.1  | 35.7  | 38.2  | 39.8  | 38.8  | 37.0  | 37.5  | 34.2  | 33.6  | 33.8  | 33.9  | 33.6 |
|                                                                                  |             | MCWB   | 23.5  | 24.2  | 25.3  | 27.4  | 27.3  | 27.4  | 27.2  | 27.4  | 26.9  | 26.7  | 24.6  | 24.9 |
|                                                                                  | 2%          | DB     | 32.1  | 34.4  | 36.9  | 38.3  | 37.0  | 35.4  | 34.8  | 33.1  | 32.7  | 32.9  | 32.5  | 32.2 |
|                                                                                  |             | MCWB   | 22.8  | 24.2  | 25.0  | 26.6  | 27.0  | 26.9  | 26.8  | 27.3  | 26.9  | 26.2  | 24.6  | 24.0 |
|                                                                                  | 5%          | DB     | 31.1  | 33.3  | 35.6  | 37.1  | 35.7  | 34.3  | 33.3  | 32.2  | 32.0  | 32.1  | 31.7  | 31.0 |
|                                                                                  |             | MCWB   | 22.3  | 23.8  | 25.2  | 26.5  | 26.9  | 26.9  | 26.7  | 27.0  | 26.7  | 25.7  | 24.4  | 23.5 |
|                                                                                  | 10%         | DB     | 29.9  | 31.8  | 34.3  | 35.7  | 34.3  | 33.3  | 32.0  | 31.3  | 31.1  | 31.4  | 30.8  | 29.7 |
|                                                                                  |             | MCWB   | 21.7  | 23.2  | 25.3  | 26.4  | 26.9  | 27.0  | 26.6  | 26.8  | 26.7  | 25.4  | 24.0  | 22.5 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 25.5  | 26.2  | 27.6  | 29.1  | 28.7  | 28.6  | 28.2  | 28.4  | 28.2  | 27.6  | 26.7  | 25.7 |
|                                                                                  |             | MCDB   | 30.3  | 31.6  | 32.9  | 35.7  | 33.6  | 33.1  | 32.5  | 31.9  | 31.5  | 31.7  | 30.9  | 32.0 |
|                                                                                  | 2%          | WB     | 24.6  | 25.5  | 26.9  | 28.0  | 28.1  | 28.0  | 27.6  | 27.9  | 27.7  | 27.0  | 26.1  | 24.9 |
|                                                                                  |             | MCDB   | 29.1  | 31.1  | 32.8  | 34.6  | 33.4  | 32.6  | 32.0  | 31.4  | 30.8  | 30.9  | 30.2  | 30.4 |
|                                                                                  | 5%          | WB     | 23.7  | 25.0  | 26.4  | 27.4  | 27.7  | 27.6  | 27.2  | 27.5  | 27.4  | 26.6  | 25.5  | 24.1 |
|                                                                                  |             | MCDB   | 28.7  | 30.7  | 32.6  | 34.1  | 33.0  | 32.2  | 31.4  | 30.9  | 30.4  | 30.2  | 29.4  | 29.2 |
|                                                                                  | 10%         | WB     | 22.9  | 24.4  | 25.9  | 26.9  | 27.3  | 27.3  | 27.0  | 27.1  | 27.1  | 26.2  | 24.9  | 23.4 |
|                                                                                  |             | MCDB   | 28.0  | 29.8  | 32.3  | 33.6  | 32.5  | 31.7  | 30.9  | 30.4  | 29.9  | 29.5  | 28.5  | 28.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.2  | 12.0  | 11.2  | 10.6  | 8.8   | 7.3   | 6.4   | 6.1   | 6.1   | 7.8   | 9.6   | 11.2 |
|                                                                                  |             | MCDBR  | 12.5  | 12.6  | 12.8  | 12.0  | 10.4  | 8.8   | 8.4   | 7.3   | 7.3   | 8.4   | 10.2  | 11.7 |
|                                                                                  |             | MCWBR  | 4.7   | 4.2   | 3.8   | 3.5   | 2.9   | 2.6   | 2.6   | 2.8   | 2.9   | 3.1   | 3.6   | 4.5  |
|                                                                                  | 5% WB       | MCDBR  | 10.6  | 11.1  | 11.5  | 11.3  | 9.3   | 7.8   | 6.9   | 6.8   | 6.7   | 7.5   | 8.6   | 10.9 |
|                                                                                  |             | MCWBR  | 4.7   | 3.9   | 4.0   | 3.6   | 3.3   | 2.9   | 2.7   | 2.9   | 2.9   | 3.0   | 3.4   | 4.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.475  | 0.592 | 0.664 | 0.624 | 0.516 | 0.471 | 0.459 | 0.465 | 0.508 | 0.527 | 0.470 | 0.466 |      |
|                                                                                  | taud        | 2.090  | 1.797 | 1.679 | 1.808 | 2.131 | 2.279 | 2.312 | 2.284 | 2.150 | 2.062 | 2.196 | 2.142 |      |
|                                                                                  | Ebn at Noon | 809    | 731   | 694   | 724   | 796   | 825   | 836   | 838   | 804   | 776   | 809   | 805   |      |
|                                                                                  | Edh at Noon | 155    | 218   | 252   | 222   | 158   | 135   | 131   | 136   | 155   | 165   | 139   | 144   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 4.90   | 5.18  | 5.29  | 5.71  | 5.52  | 5.15  | 4.80  | 4.70  | 4.61  | 5.03  | 4.93  | 4.73  |      |
|                                                                                  | RadStd      | 0.28   | 0.26  | 0.24  | 0.26  | 0.30  | 0.27  | 0.36  | 0.30  | 0.34  | 0.34  | 0.17  | 0.12  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (21 neighbors) | +0.28 | N/A     | N/A    | N/A     | +0.13 | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page